

Dev Patel

Senior Software Engineer • Cloud Platform • Event-Driven Systems • AI / Agent Platforms
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SUMMARY

Cloud-platform engineer with 5 years building event-driven systems for high-volume dental manufacturing. Full-stack owner across a portfolio of manufacturing platforms — architecting and deploying the AWS infrastructure other engineers build on, and defining the Terraform, serverless, and authentication patterns now standard across 40+ services and 13+ applications. Recent work spans real-time IoT telemetry (1M+ daily events, sub-second visibility) and a production AI agent platform on Amazon Bedrock.

EXPERIENCE

Glidewell Dental — Software Engineer III — Irvine, CA

Sep 2021 – Present

- First software engineer on the Automation team, joining mechanical and controls engineers building factory lines and piloting IoT on the shop floor — independently built the AWS telemetry backend that ingested their device data and the full software platform around it, from requirements through production. It became the foundation for the team's later systems.
- Own the full stack across every project on the team: architect and deploy all AWS infrastructure, then guide the engineers building on top of it through design, implementation, and review before each change reaches production. Standardized on serverless-first architecture, so cost tracks real floor activity, not provisioned capacity.
- Established the standardized Terraform + Lambda delivery pattern — shared modules and GitLab CI that builds and deploys on apply, with multi-account promotion — now the default across 40+ backend services, cutting new-service setup from days to hours.
- Built the Manufacturing Execution System (MES), digitizing physical, flow-based production processes so every step emits structured data — then transforming it into real-time KPIs and AI-generated charts for plant leadership.
- Built a production AI agent platform on Amazon Bedrock: a planner-routed, multi-model (Claude + Amazon Nova) tool-use agent with live DynamoDB access, a sandboxed code interpreter, and NL-to-SQL. It structures free-text ticket narratives into queryable records, builds charts and KPIs on demand instead of fixed dashboards, and guides staff in-product — eliminating the week of cross-team investigation a single customer concern used to require.
- Architected AWS IoT and Kafka/MSK pipelines processing 1M+ daily telemetry events, cutting operational-visibility latency from minutes to sub-second and giving production facilities real-time fault visibility for the first time.
- Designed an IoT device-management and OTA platform for CNC milling machines: zero-touch X.509 provisioning, least-privilege per-device policies, over-the-air firmware/config delivery, and a device-shadow reconciliation engine with hash-verified config, conflict detection, and automated recovery.
- Delivered the serverless analytics behind a 3,500+ machine milling-fleet dashboard — idle-machine detection, an Entity Hub 360 view, and Zendesk / Pega field-service integrations — unifying five facilities and nine production lines and eliminating hours of manual status checks daily.
- Built a centralized authentication layer: one reusable Lambda authorizer validating dual Cognito and Azure AD (MSAL) tokens plus API keys across HTTP, REST, and WebSocket APIs, standardizing secure access across internal applications.

SELECTED SYSTEMS

AI Agent Platform — Manufacturing Execution System. Amazon Bedrock (Claude, Nova), Node.js, DynamoDB, API Gateway WebSockets. Planner-routed, multi-model agent with a 10-tool registry — DynamoDB access, a sandboxed code interpreter, analytics presets, and email/SNS/IoT actions — answering natural-language questions over live production data.

Unified Real-Time Data Exchange Platform. Python, AWS IoT Core, WebSocket API, DynamoDB, Timestream for InfluxDB, Terraform. Topic-routed pub/sub backbone ingesting all factory MQTT traffic and fanning out to webhooks, live WebSocket clients, and a time-series store, with a no-code message transformer and HIPAA-oriented PII hashing.

3D Mesh QC Viewer. Next.js, React, three.js. In-browser STL / PLY / OBJ rendering with reference-vs-test geometric deviation heatmaps and PDF/zip export for dental-part quality control.

TECHNICAL SKILLS

Languages: Python, JavaScript / Node.js, SQL, Java

AWS: Lambda, IoT Core, Greengrass, Kinesis, Kafka / MSK, DynamoDB, Timestream, S3, SQS / SNS / SES, API Gateway, VPC, Amplify

AI & ML: Amazon Bedrock (Claude, Nova), agent orchestration & tool-use, NL-to-SQL over operational data, prompt engineering, production LLM integration

Frontend: React, MUI, Next.js, Angular, Highcharts / Plotly, three.js

Infrastructure & Security: Terraform, Kubernetes, GitLab CI/CD, Jenkins / CloudBees, Git; IAM, Cognito, Azure AD / MSAL; Azure, GCP

Enterprise Integrations: Zendesk, Pega, D365 / Dataverse

EDUCATION

Rutgers University — B.S. in Computer Science, New Brunswick, NJ

May 2021